

R CODE CLASS ACTIVITY -15/7/26

```
1) import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# -----
# Generate Sample Dataset
# -----
np.random.seed(10)

n = 150

data = pd.DataFrame({
    "Sensor": range(1, n+1),
    "PM2.5": np.random.randint(10, 250, n),
    "NO2": np.random.randint(5, 180, n),
    "CO": np.random.uniform(0.1, 5.0, n),
    "Temperature": np.random.randint(20, 45, n),
    "Humidity": np.random.randint(30, 95, n),
    "X": np.random.randint(1, 21, n),
    "Y": np.random.randint(1, 21, n)
})

print(data.head())

# -----
# Sequential Color Palette
# -----
plt.figure(figsize=(8,6))
plt.scatter(data["X"], data["Y"],
            c=data["PM2.5"],
            cmap="Greens",
            s=120)

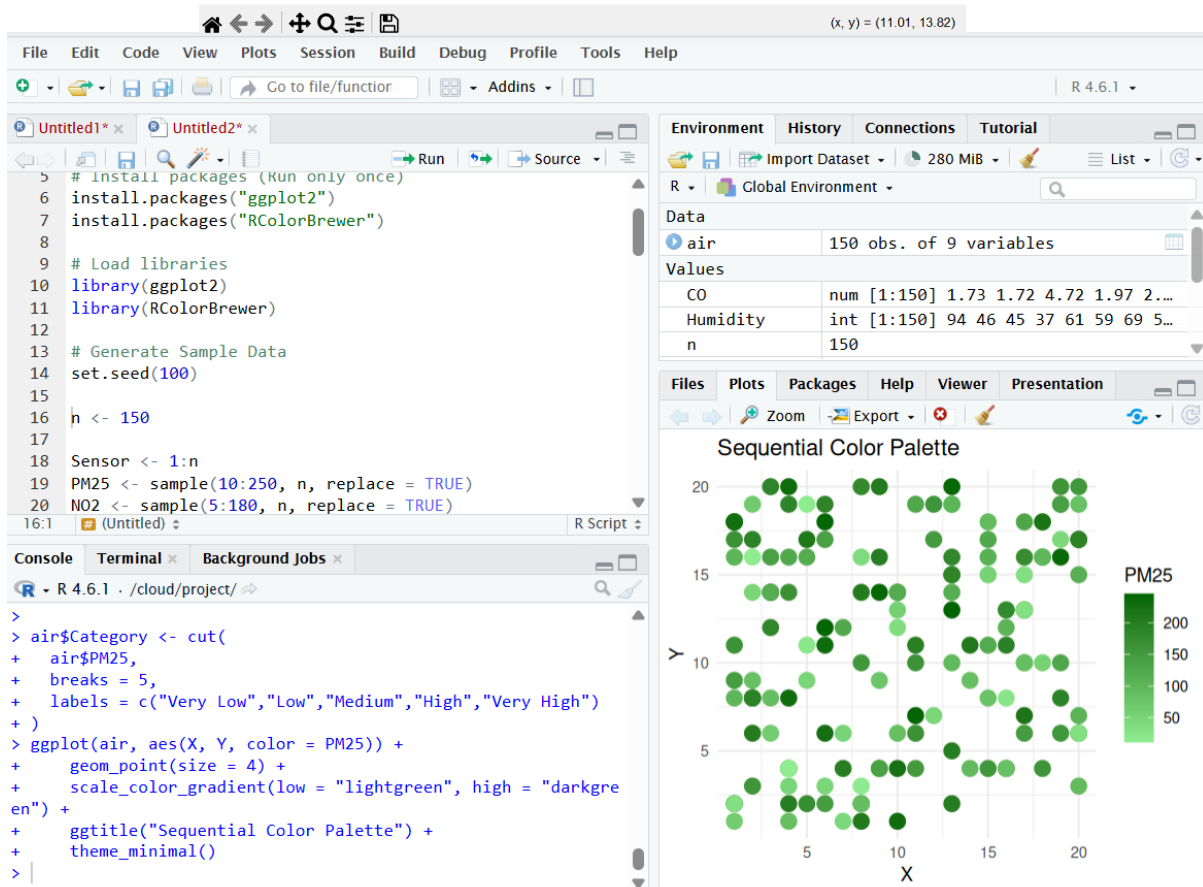
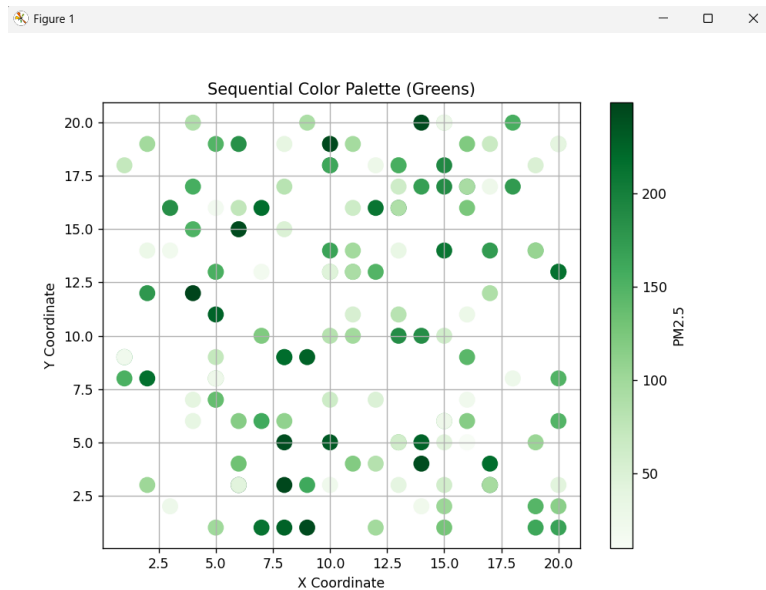
plt.colorbar(label="PM2.5")
plt.title("Sequential Color Palette (Greens)")
plt.xlabel("X Coordinate")
plt.ylabel("Y Coordinate")
plt.grid(True)
plt.show()

# -----
# Diverging Color Palette
```

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```
# -----  
plt.figure(figsize=(8,6))  
plt.scatter(data["X"], data["Y"],  
            c=data["PM2.5"],  
            cmap="coolwarm",  
            s=120)  
  
plt.colorbar(label="PM2.5")  
plt.title("Diverging Color Palette (Coolwarm)")  
plt.xlabel("X Coordinate")  
plt.ylabel("Y Coordinate")  
plt.grid(True)  
plt.show()  
  
# -----  
# Qualitative Color Palette  
# -----  
  
colors = sns.color_palette("Set2", 5)  
  
category = pd.cut(data["PM2.5"],  
                  bins=5,  
                  labels=False)  
  
plt.figure(figsize=(8,6))  
  
for i in range(5):  
    subset = data[category == i]  
    plt.scatter(subset["X"],  
                subset["Y"],  
                color=colors[i],  
                s=120,  
                label=f"Group {i+1}")  
  
plt.title("Qualitative Palette (Set2)")  
plt.xlabel("X Coordinate")  
plt.ylabel("Y Coordinate")  
plt.legend()  
plt.grid(True)  
plt.show()
```

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2)

```
#=====
```

```
# EXPERIMENT 2
```

```
# Earthquake Risk Analysis
```

```
#=====
```

```
# Install package (Run only once)
```

```
if (!require(ggplot2)) {
```

```
  install.packages("ggplot2")
```

```
  library(ggplot2)
```

```
} else {
```

```
  library(ggplot2)
```

```
}
```

```
#-----
```

```
# Generate Sample Earthquake Data
```

```
#-----
```

```
set.seed(123)
```

```
n <- 1000
```

```
Magnitude <- pmin(rexp(n, rate = 1/1.5) + 0.5, 9.5)
```

```
Depth <- sample(1:700, n, replace = TRUE)
```

```
Latitude <- runif(n, -90, 90)
```

```
Longitude <- runif(n, -180, 180)
```

```
Aftershock <- sample(0:60, n, replace = TRUE)
```

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```
earthquake <- data.frame(
```

```
  Magnitude,
```

```
  Depth,
```

```
  Latitude,
```

```
  Longitude,
```

```
  Aftershock
```

```
)
```

```
print(head(earthquake))
```

```
#=====
```

```
# 1. Scatter Plot - Linear Scale
```

```
#=====
```

```
p1 <- ggplot(earthquake,
```

```
  aes(x = Magnitude,
```

```
    y = Depth,
```

```
    color = Aftershock)) +
```

```
geom_point(size = 2) +
```

```
labs(
```

```
  title = "Earthquake Distribution - Linear Scale",
```

```
  x = "Magnitude",
```

```
  y = "Depth (km)"
```

```
) +
```

```
theme_minimal()
```

```
print(p1)
```

```
#=====
```

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2. Scatter Plot - Log Scale

#=====

```
p2 <- ggplot(earthquake,
             aes(x = Magnitude,
                 y = Depth,
                 color = Aftershock)) +
  geom_point(size = 2) +
  scale_y_log10() +
  labs(
    title = "Earthquake Distribution - Log Scale",
    x = "Magnitude",
    y = "Depth (Log Scale)"
  ) +
  theme_minimal()

print(p2)
```

#=====

3. Histogram - Linear Scale

#=====

```
p3 <- ggplot(earthquake,
             aes(x = Magnitude)) +
  geom_histogram(
    bins = 30,
    fill = "skyblue",
    color = "black"
  ) +
```

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```
labs(  
  title = "Magnitude Distribution - Linear",  
  x = "Magnitude",  
  y = "Frequency"  
) +  
theme_minimal()
```

```
print(p3)
```

```
#=====
```

4. Histogram - Log Scale

```
#=====
```

```
p4 <- ggplot(earthquake,  
  aes(x = Magnitude)) +  
geom_histogram(  
  bins = 30,  
  fill = "orange",  
  color = "black"  
) +  
scale_y_log10() +  
labs(  
  title = "Magnitude Distribution - Log Scale",  
  x = "Magnitude",  
  y = "Frequency (Log)"  
) +  
theme_minimal()
```

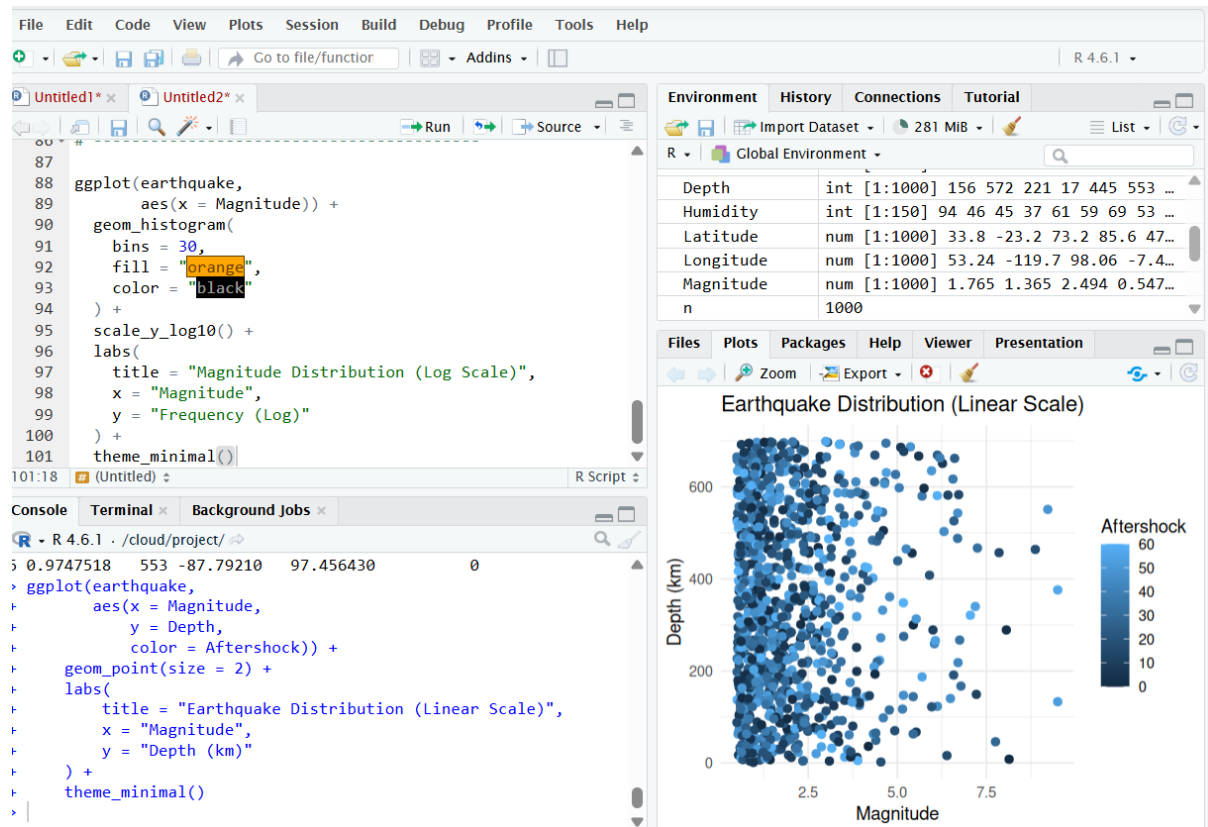
```
print(p4)
```

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```
#=====
```

```
# End of Program
```

```
#=====
```



3)

```
#=====
```

```
# EXPERIMENT 3
```

```
# Wind Direction Analysis for Airport Operations
```

```
#=====
```

```
# Install packages (Run only once)
```

```
if (!require(ggplot2)) install.packages("ggplot2")
```

```
library(ggplot2)
```

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```
#-----
```

```
# Generate Sample Data
```

```
#-----
```

```
set.seed(123)
```

```
n <- 1000
```

```
WindSpeed <- runif(n, 0, 25)
```

```
WindDirection <- runif(n, 0, 360)
```

```
wind <- data.frame(
```

```
  WindSpeed,
```

```
  WindDirection
```

```
)
```

```
print(head(wind))
```

```
#=====
```

```
# 1. Cartesian Scatter Plot
```

```
#=====
```

```
p1 <- ggplot(wind,
```

```
  aes(x = 1:n,
```

```
    y = WindDirection,
```

```
    color = WindSpeed)) +
```

```
geom_point(size = 2) +
```

```
labs(
```

```
  title = "Wind Direction - Cartesian Plot",
```

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```
x = "Observation",
y = "Wind Direction (Degrees)"
) +
theme_minimal()

print(p1)

#=====

# 2. Polar Coordinate Plot

#=====

p2 <- ggplot(wind,
             aes(x = factor(round(WindDirection/30)*30),
                 fill = WindSpeed)) +
  geom_bar() +
  coord_polar() +
  labs(
    title = "Wind Rose (Polar Coordinate Plot)",
    x = "Direction",
    y = "Frequency"
  ) +
  theme_minimal()

print(p2)

#=====

# 3. Wind Speed Histogram

#=====
```

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```
p3 <- ggplot(wind,  
             aes(x = WindSpeed)) +  
  geom_histogram(  
    bins = 20,  
    fill = "skyblue",  
    color = "black"  
  ) +  
  labs(  
    title = "Wind Speed Distribution",  
    x = "Wind Speed (m/s)",  
    y = "Frequency"  
  ) +  
  theme_minimal()
```

```
print(p3)
```

```
#=====
```

```
# 4. Wind Direction Histogram
```

```
#=====
```

```
p4 <- ggplot(wind,  
             aes(x = WindDirection)) +  
  geom_histogram(  
    bins = 12,  
    fill = "orange",  
    color = "black"  
  ) +  
  labs(  
    title = "Wind Direction Distribution",
```

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```
x = "Direction (Degrees)",
```

```
y = "Frequency"
```

```
) +
```

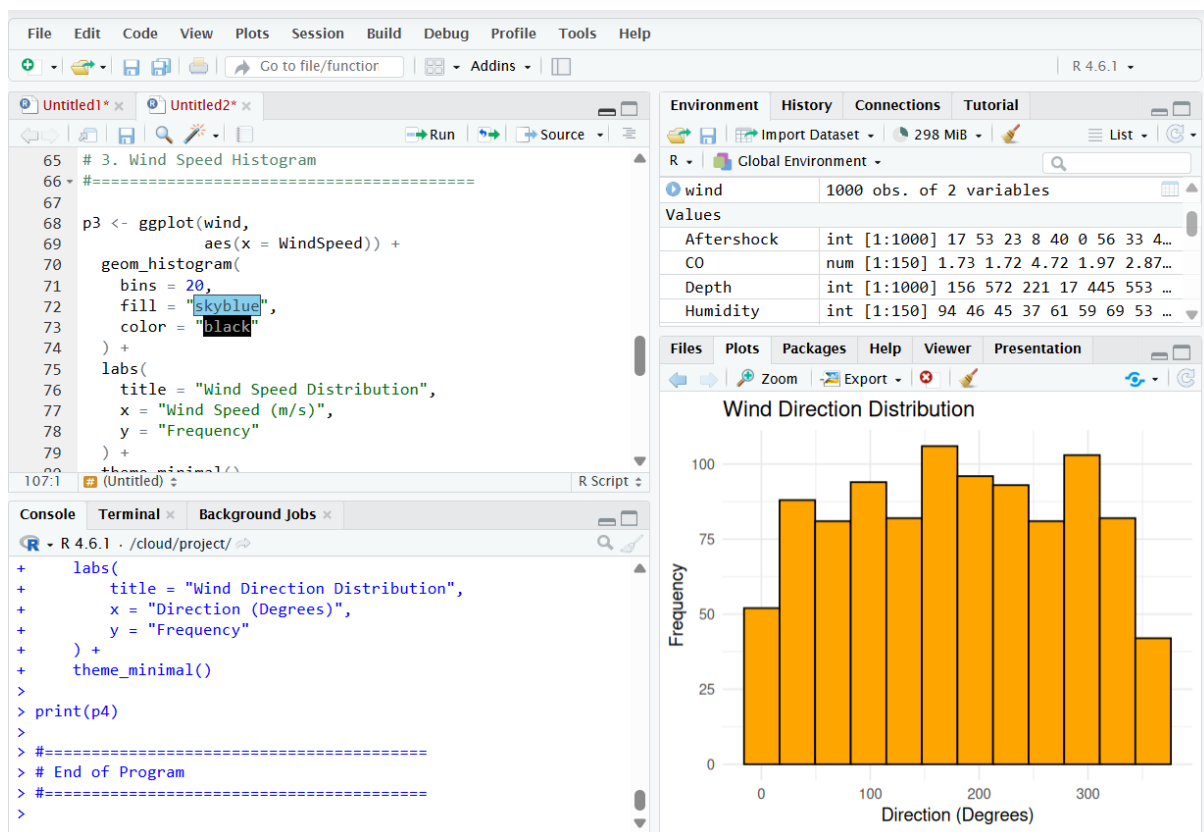
```
theme_minimal()
```

```
print(p4)
```

```
#=====
```

```
# End of Program
```

```
#=====
```



4)

```
#=====
```

```
# EXPERIMENT 4
```

```
# Hospital Patient Monitoring Dashboard
```

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```
#=====
```

```
# Install packages (Run only once)
```

```
if (!require(ggplot2)) install.packages("ggplot2")
```

```
library(ggplot2)
```

```
#-----
```

```
# Generate Sample Patient Data
```

```
#-----
```

```
set.seed(100)
```

```
n <- 200
```

```
Time <- 1:n
```

```
HeartRate <- sample(50:140, n, replace = TRUE)
```

```
Oxygen <- sample(85:100, n, replace = TRUE)
```

```
SystolicBP <- sample(90:180, n, replace = TRUE)
```

```
Respiration <- sample(10:35, n, replace = TRUE)
```

```
Temperature <- round(runif(n, 35, 40),1)
```

```
patient <- data.frame(
```

```
  Time,
```

```
  HeartRate,
```

```
  Oxygen,
```

```
  SystolicBP,
```

```
  Respiration,
```

```
  Temperature
```

```
)
```

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```
print(head(patient))
```

```
#=====
```

```
# Heart Rate Dashboard
```

```
#=====
```

```
patient$HR_Status <- ifelse(patient$HeartRate < 60 | patient$HeartRate > 100,  
                             "Abnormal","Normal")
```

```
p1 <- ggplot(patient,  
              aes(Time, HeartRate, color = HR_Status)) +  
  geom_point(size = 2) +  
  scale_color_manual(values = c("Normal"="green",  
                                "Abnormal"="red")) +  
  ggtitle("Heart Rate Monitoring") +  
  theme_minimal()
```

```
print(p1)
```

```
#=====
```

```
# Oxygen Saturation Dashboard
```

```
#=====
```

```
patient$O2_Status <- ifelse(patient$Oxygen < 95,  
                             "Low","Normal")
```

```
p2 <- ggplot(patient,  
              aes(Time, Oxygen, color = O2_Status)) +
```

R CODE CLASS ACTIVITY -15/7/26

```
geom_point(size = 2) +  
scale_color_manual(values = c("Normal"="blue",  
                              "Low"="red")) +  
ggtitle("Oxygen Saturation") +  
theme_minimal()
```

```
print(p2)
```

```
#=====
```

```
# Blood Pressure Dashboard
```

```
#=====
```

```
patient$BP_Status <- ifelse(patient$SystolicBP > 140,  
                             "High","Normal")
```

```
p3 <- ggplot(patient,  
              aes(Time, SystolicBP, color = BP_Status)) +  
  geom_line() +  
  scale_color_manual(values = c("Normal"="green",  
                                "High"="red")) +  
  ggtitle("Blood Pressure") +  
  theme_minimal()
```

```
print(p3)
```

```
#=====
```

```
# Temperature Dashboard
```

```
#=====
```

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```
patient$Temp_Status <- ifelse(patient$Temperature > 37.5,  
                              "High","Normal")
```

```
p4 <- ggplot(patient,  
             aes(Time, Temperature, color = Temp_Status)) +  
  geom_line() +  
  scale_color_manual(values = c("Normal"="blue",  
                                "High"="red")) +  
  ggtitle("Body Temperature") +  
  theme_minimal()
```

```
print(p4)
```

```
#=====
```

Respiration Dashboard

```
#=====
```

```
patient$Resp_Status <- ifelse(patient$Respiration > 20,  
                              "High","Normal")
```

```
p5 <- ggplot(patient,  
             aes(Time, Respiration, color = Resp_Status)) +  
  geom_line() +  
  scale_color_manual(values = c("Normal"="green",  
                                "High"="red")) +  
  ggtitle("Respiration Rate") +  
  theme_minimal()
```

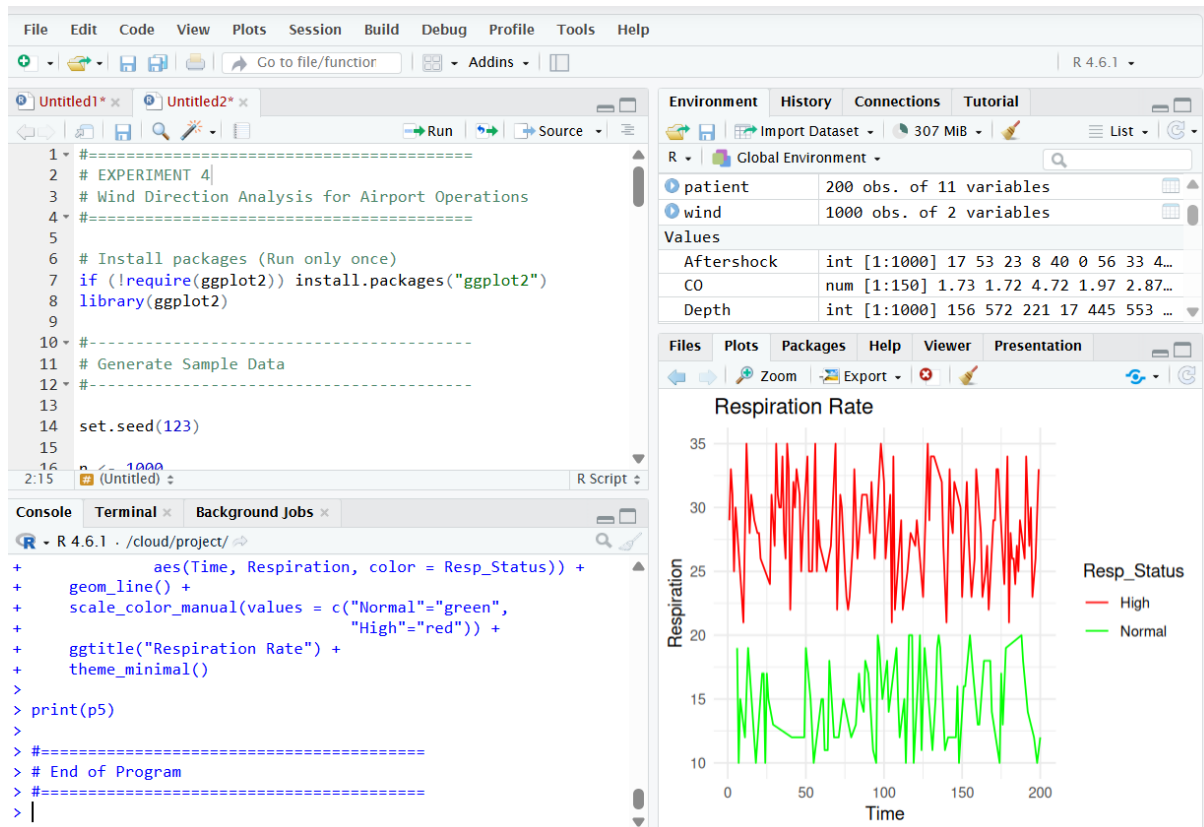
```
print(p5)
```

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#=====

End of Program

#=====



5)

#=====

EXPERIMENT 5

Global Climate Change Study

#=====

Install packages (Run only once)

if (!require(ggplot2)) install.packages("ggplot2")

if (!require(RColorBrewer)) install.packages("RColorBrewer")

library(ggplot2)

R CODE CLASS ACTIVITY -15/7/26

```
library(RColorBrewer)
```

```
#-----
```

```
# Generate Sample Data
```

```
#-----
```

```
set.seed(123)
```

```
countries <- paste("Country",1:180)
```

```
years <- 1980:2025
```

```
climate <- expand.grid(
```

```
  Country = countries,
```

```
  Year = years
```

```
)
```

```
climate$Temperature_Anomaly <- round(runif(
```

```
  nrow(climate),
```

```
  -3,
```

```
  3
```

```
),2)
```

```
head(climate)
```

```
#=====
```

```
# Heatmap 1 - Blue White Red
```

```
#=====
```

```
p1 <- ggplot(climate,
```

R CODE CLASS ACTIVITY -15/7/26

```
      aes(x=Year,
          y=Country,
          fill=Temperature_Anomaly))+
geom_tile()+
scale_fill_gradient2(
  low="blue",
  mid="white",
  high="red",
  midpoint=0
)+
ggtitle("Temperature Anomalies - Blue White Red")+
theme_minimal()

print(p1)

#=====

# Heatmap 2 - Spectral Palette

#=====

p2 <- ggplot(climate,
  aes(x=Year,
      y=Country,
      fill=Temperature_Anomaly))+
geom_tile()+
scale_fill_distiller(
  palette="Spectral"
)+
ggtitle("Temperature Anomalies - Spectral")+
theme_minimal()
```

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```
print(p2)
```

```
#=====
```

```
# Heatmap 3 - RdBu Palette
```

```
#=====
```

```
p3 <- ggplot(climate,  
             aes(x=Year,  
                 y=Country,  
                 fill=Temperature_Anomaly))+  
geom_tile()+  
scale_fill_distiller(  
  palette="RdBu",  
  direction=-1  
)+  
ggtitle("Temperature Anomalies - RdBu")+  
theme_minimal()
```

```
print(p3)
```

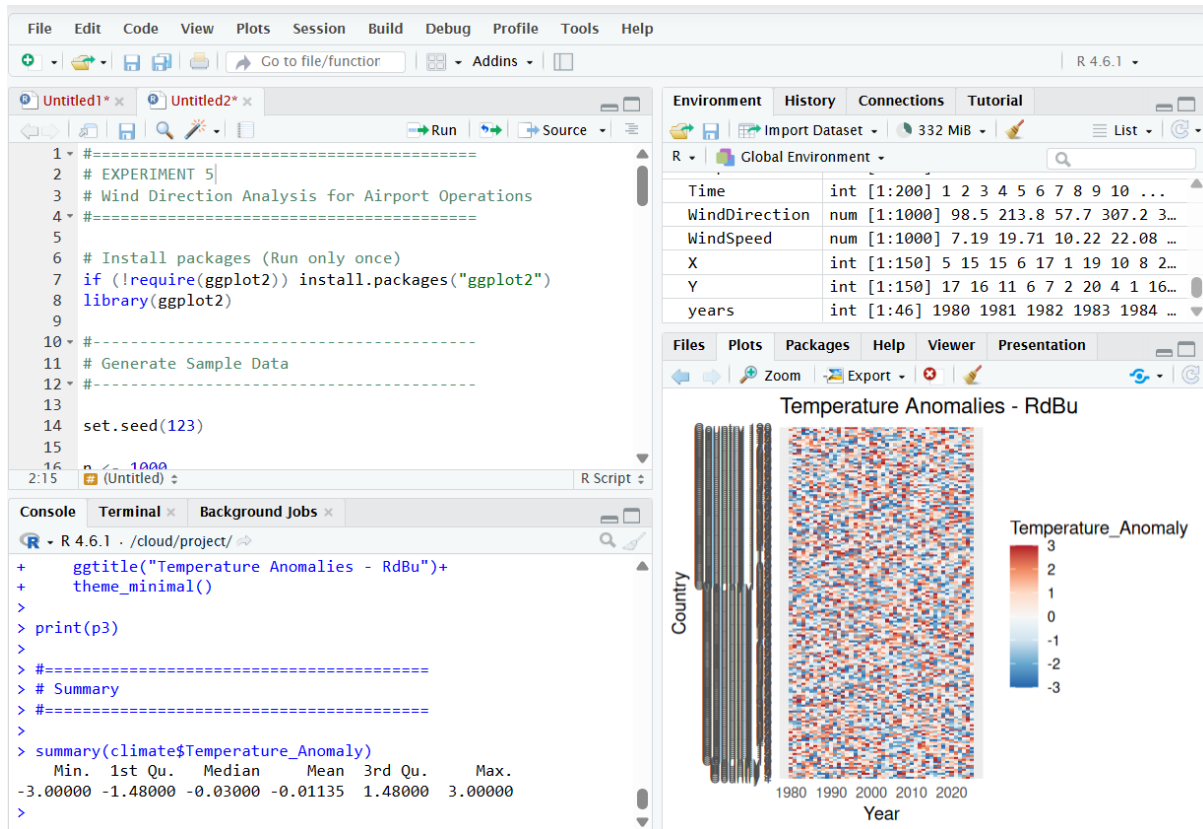
```
#=====
```

```
# Summary
```

```
#=====
```

R CODE CLASS ACTIVITY -15/7/26

summary(climate\$Temperature_Anomaly)



6)

#=====

EXPERIMENT 6

E-Commerce Sales Analytics

#=====

Install packages (Run only once)

if (!require(ggplot2)) install.packages("ggplot2")

library(ggplot2)

#-----

Generate Sample Data

#-----

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```
set.seed(123)
```

```
Category <- paste("Category", 1:50)
```

```
Region <- paste("Region", 1:30)
```

```
sales <- expand.grid(
```

```
  Category = Category,
```

```
  Region = Region
```

```
)
```

```
sales$Sales <- sample(1000:100000,
```

```
  nrow(sales),
```

```
  replace = TRUE)
```

```
head(sales)
```

```
#=====
```

```
# 1. Bubble Chart
```

```
#=====
```

```
p1 <- ggplot(sales,
```

```
  aes(x = Category,
```

```
    y = Region,
```

```
    size = Sales,
```

```
    color = Category)) +
```

```
geom_point(alpha = 0.8) +
```

```
ggtitle("Bubble Chart - Sales by Category and Region") +
```

```
theme_minimal() +
```

```
theme(axis.text.x = element_text(angle = 90))
```

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```
print(p1)
```

```
#=====
```

```
# 2. Heatmap
```

```
#=====
```

```
p2 <- ggplot(sales,  
             aes(x = Category,  
                 y = Region,  
                 fill = Sales)) +  
  geom_tile() +  
  scale_fill_gradient(low = "yellow",  
                      high = "red") +  
  ggtitle("Heatmap - Sales Volume") +  
  theme_minimal() +  
  theme(axis.text.x = element_text(angle = 90))
```

```
print(p2)
```

```
#=====
```

```
# 3. Bar Chart
```

```
#=====
```

```
total_sales <- aggregate(Sales ~ Category,  
                          sales,  
                          sum)
```

```
p3 <- ggplot(total_sales,  
             aes(x = reorder(Category, Sales),  
                 y = Sales,
```

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```
      fill = Sales)) +  
geom_col() +  
coord_flip() +  
scale_fill_gradient(low = "lightblue",  
                    high = "darkblue") +  
ggtitle("Total Sales by Category") +  
theme_minimal()  
  
print(p3)  
  
#=====
```

4. Scatter Plot

```
#=====
```



```
p4 <- ggplot(sales,  
            aes(x = Sales,  
                y = Region,  
                color = Category)) +  
geom_point(size = 2) +  
ggtitle("Sales Distribution") +  
theme_minimal()  
  
print(p4)  
  
#=====
```

End of Program

```
#=====
```

The screenshot displays the RStudio environment with the following components:

- Menu Bar:** File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, Help.
- Toolbar:** Includes icons for file operations (save, open, print), running code, and source control.
- Script Editor:** Contains an R script with comments and code for installing packages and setting a seed.


```

1 #=====
2 # EXPERIMENT 6
3 # Wind Direction Analysis for Airport Operations
4 #=====
5
6 # Install packages (Run only once)
7 if(!require(ggplot2)) install.packages("ggplot2")
8 library(ggplot2)
9
10 #-----
11 # Generate Sample Data
12 #-----
13
14 set.seed(123)
15
16 n <- 1000
17
18 # (Untitled) R Script
      
```
- Console:** Shows the execution of the script, including the installation of the ggplot2 package.


```

> R 4.6.1 . /cloud/project/
+ aes(x = Sales,
+     y = Region,
+     color = Category)) +
+   geom_point(size = 2) +
+   ggtitle("Sales Distribution") +
+   theme_minimal()
>
> print(p4)
>
> #=====
> # End of Program
> #=====
>
      
```
- Plots Panel:** Displays a scatter plot titled "Sales Distribution". The x-axis is labeled "Sales" and the y-axis is labeled "Region". The plot shows a distribution of points colored by "Category", ranging from Category 1 to Category 50. The plot is styled with a white background and a black border.
- Environment Panel:** Shows the current environment with variables like "total_sales" and "wind".
- Files Panel:** Displays the file structure of the project, including a "Region" folder and a "Sales" file.

#=====

Satellite Image Vegetation Analysis

#=====

```
if (!require(ggplot2)) install.packages("ggplot2")
```

#-----

#-----

```
set.seed(123)
```

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```
n <- 1000
```

```
X <- sample(1:100, n, replace = TRUE)
```

```
Y <- sample(1:100, n, replace = TRUE)
```

```
NDVI <- round(runif(n, -1, 1), 2)
```

```
vegetation <- data.frame(  
  X,  
  Y,  
  NDVI  
)
```

```
head(vegetation)
```

```
#=====
```

```
# 1. Sequential Green Palette
```

```
#=====
```

```
p1 <- ggplot(vegetation,  
  aes(x = X,  
    y = Y,  
    color = NDVI)) +  
  geom_point(size = 3) +  
  scale_color_gradient(  
    low = "yellow",  
    high = "darkgreen"  
  ) +  
  ggtitle("Sequential Palette - Vegetation Health") +
```

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```
theme_minimal()
```

```
print(p1)
```

```
#=====
```

```
# 2. Diverging Palette
```

```
#=====
```

```
p2 <- ggplot(vegetation,  
             aes(x = X,  
                 y = Y,  
                 color = NDVI)) +  
  geom_point(size = 3) +  
  scale_color_gradient2(  
    low = "blue",  
    mid = "yellow",  
    high = "darkgreen",  
    midpoint = 0  
  ) +  
  ggtitle("Diverging Palette - NDVI") +  
  theme_minimal()
```

```
print(p2)
```

```
#=====
```

```
# 3. NDVI Histogram
```

```
#=====
```

```
p3 <- ggplot(vegetation,  
             aes(x = NDVI)) +
```

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```
geom_histogram(  
  bins = 25,  
  fill = "forestgreen",  
  color = "black"  
) +  
ggtitle("NDVI Distribution") +  
theme_minimal()
```

```
print(p3)
```

```
#=====
```

```
# 4. NDVI Heatmap
```

```
#=====
```

```
p4 <- ggplot(vegetation,  
  aes(x = X,  
    y = Y,  
    fill = NDVI)) +  
geom_tile() +  
scale_fill_gradient2(  
  low = "blue",  
  mid = "yellow",  
  high = "darkgreen",  
  midpoint = 0  
) +  
ggtitle("NDVI Heatmap") +  
theme_minimal()
```

```
print(p4)
```

R CODE CLASS ACTIVITY -15/7/26

```
#=====
```

```
# NDVI Classification
```

```
#=====
```

```
vegetation$Class <- ifelse(
```

```
  vegetation$NDVI < 0,
```

```
  "Water",
```

```
  ifelse(
```

```
    vegetation$NDVI < 0.3,
```

```
    "Barren Land",
```

```
    "Healthy Vegetation"
```

```
  )
```

```
)
```

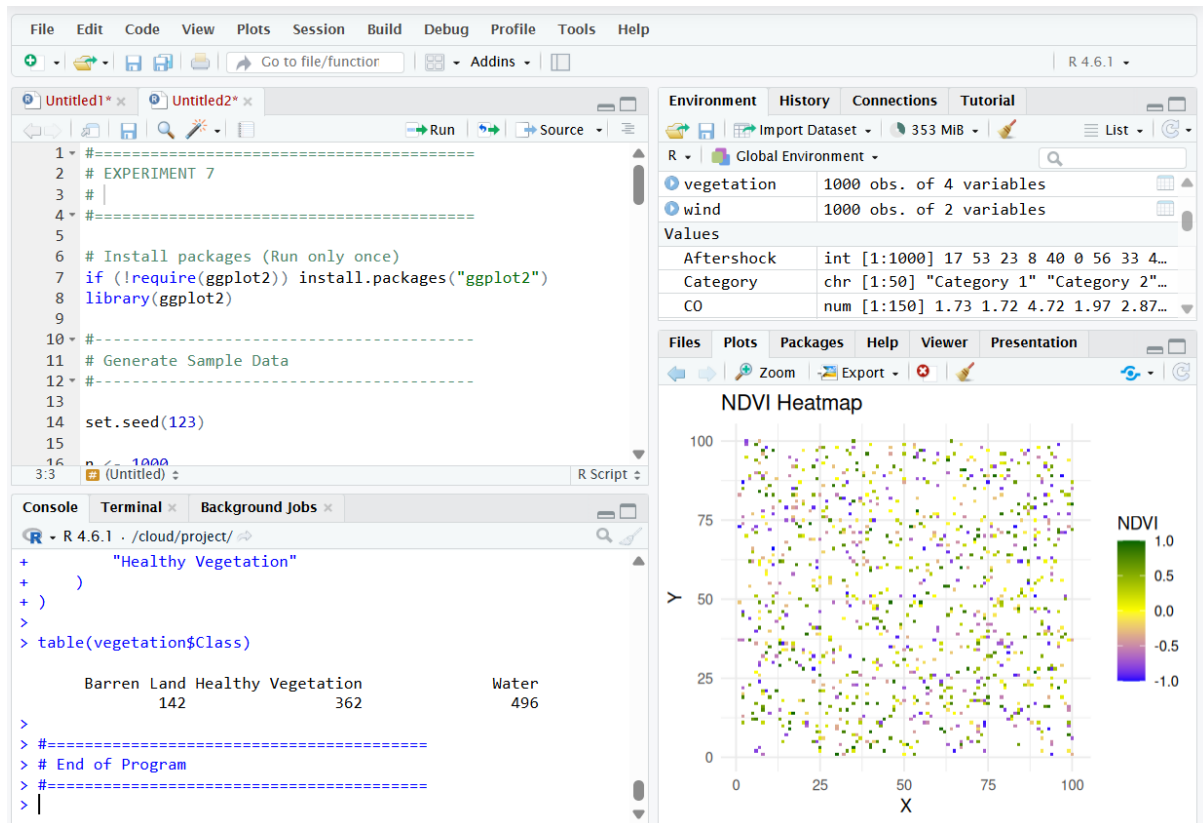
```
table(vegetation$Class)
```

```
#=====
```

```
# End of Program
```

```
#=====
```

R CODE CLASS ACTIVITY -15/7/26



8)

#=====

EXPERIMENT 8

Cryptocurrency Market Volatility

##=====

```
# Install package (Run only once)
```

```
if (!require(ggplot2)) install.packages("ggplot2")
```

```
library(ggplot2)
```

#-----

Generate Sample Cryptocurrency Data

#-----

```
set.seed(123)
```

R CODE CLASS ACTIVITY -15/7/26

```
Days <- 1:365
```

```
Bitcoin <- round(runif(365, 30000, 110000),2)
```

```
Ethereum <- round(runif(365, 1500, 7000),2)
```

```
Solana <- round(runif(365, 20, 350),2)
```

```
Cardano <- round(runif(365, 0.2, 3),2)
```

```
crypto <- data.frame(
```

```
  Days,
```

```
  Bitcoin,
```

```
  Ethereum,
```

```
  Solana,
```

```
  Cardano
```

```
)
```

```
head(crypto)
```

```
#=====
```

```
# Convert to Long Format
```

```
#=====
```

```
coins <- data.frame(
```

```
  Day = rep(Days,4),
```

```
  Price = c(Bitcoin, Ethereum, Solana, Cardano),
```

```
  Currency = rep(c("Bitcoin",
```

```
    "Ethereum",
```

```
    "Solana",
```

```
    "Cardano"),
```

```
  each = 365)
```

R CODE CLASS ACTIVITY -15/7/26

)

```
#=====
```

```
# 1. Cartesian Line Plot
```

```
#=====
```

```
p1 <- ggplot(coins,
  aes(x = Day,
      y = Price,
      color = Currency)) +
  geom_line(size = 1) +
  ggtitle("Cryptocurrency Prices (Linear Scale)") +
  xlab("Day") +
  ylab("Price (USD)") +
  theme_minimal()
```

```
print(p1)
```

```
#=====
```

```
# 2. Logarithmic Scale Plot
```

```
#=====
```

```
p2 <- ggplot(coins,
  aes(x = Day,
      y = Price,
      color = Currency)) +
  geom_line(size = 1) +
  scale_y_log10() +
  ggtitle("Cryptocurrency Prices (Log Scale)") +
  xlab("Day") +
```

R CODE CLASS ACTIVITY -15/7/26

```
ylab("Log Price") +  
theme_minimal()
```

```
print(p2)
```

```
#=====
```

```
# 3. Box Plot
```

```
#=====
```

```
p3 <- ggplot(coins,  
             aes(x = Currency,  
                 y = Price,  
                 fill = Currency)) +  
  geom_boxplot() +  
  ggtitle("Price Distribution of Cryptocurrencies") +  
  theme_minimal()
```

```
print(p3)
```

```
#=====
```

```
# 4. Histogram
```

```
#=====
```

```
p4 <- ggplot(coins,  
             aes(x = Price,  
                 fill = Currency)) +  
  geom_histogram(  
    bins = 30,  
    alpha = 0.6,  
    position = "identity"
```

R CODE CLASS ACTIVITY -15/7/26

```
) +
```

```
ggtitle("Histogram of Cryptocurrency Prices") +
```

```
theme_minimal()
```

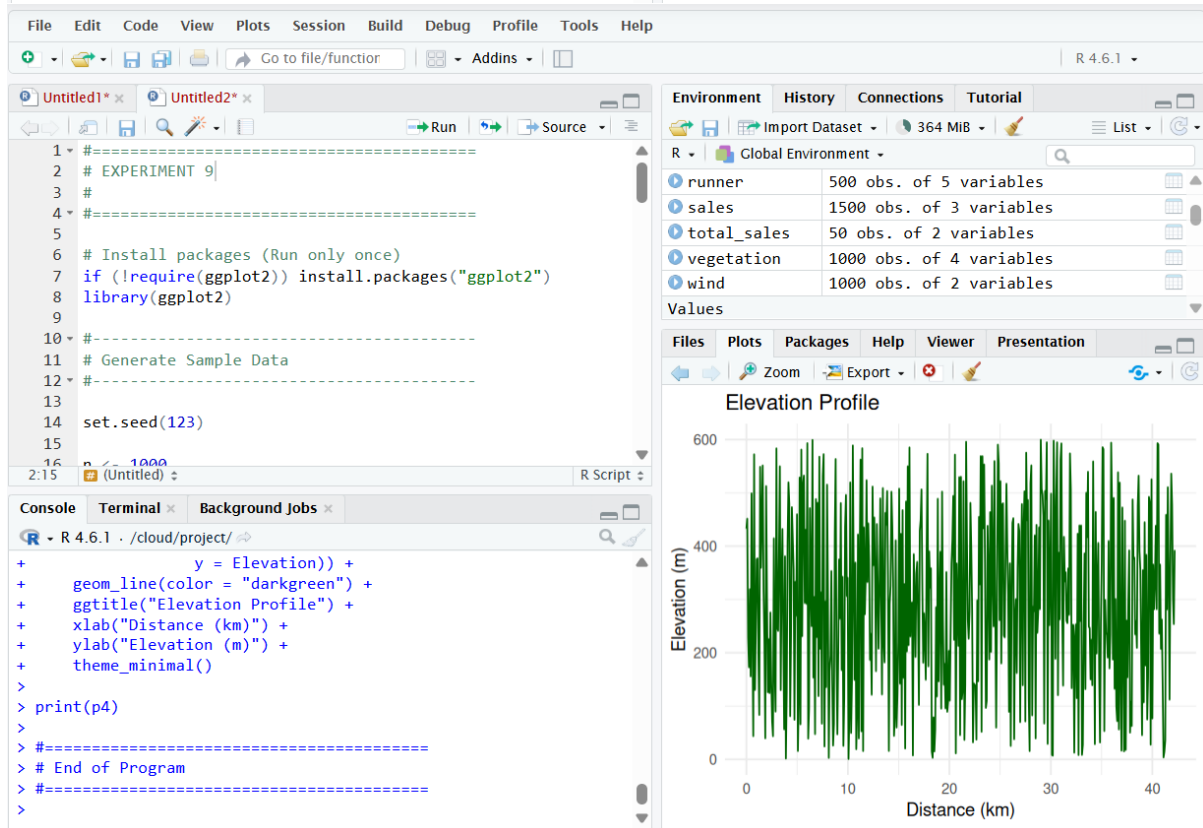
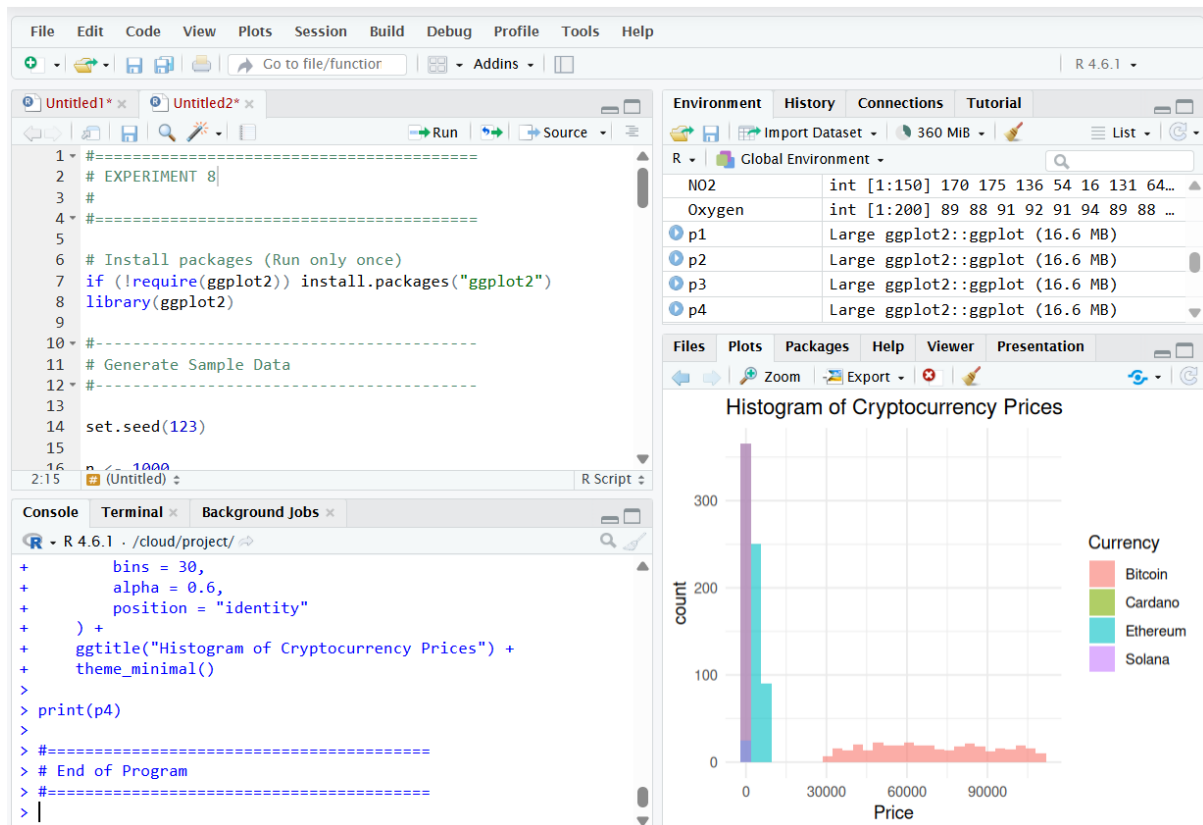
```
print(p4)
```

```
#=====
```

```
# End of Program
```

R CODE CLASS ACTIVITY -15/7/26

#=====



R CODE CLASS ACTIVITY -15/7/26

9)

```
#=====
```

```
# EXPERIMENT 9
```

```
# Marathon Runner Performance Analysis
```

```
#=====
```

```
# Install package (Run only once)
```

```
if (!require(ggplot2)) install.packages("ggplot2")
```

```
library(ggplot2)
```

```
#-----
```

```
# Generate Sample Data
```

```
#-----
```

```
set.seed(123)
```

```
n <- 500
```

```
Distance <- seq(0, 42.2, length.out = n)
```

```
Speed <- round(runif(n, 6, 18), 2)
```

```
HeartRate <- sample(90:190, n, replace = TRUE)
```

```
Elevation <- round(runif(n, 0, 600), 2)
```

```
Direction <- runif(n, 0, 360)
```

```
runner <- data.frame(
```

```
  Distance,
```

```
  Speed,
```

```
  HeartRate,
```

```
  Elevation,
```

```
  Direction
```

```
)
```

R CODE CLASS ACTIVITY -15/7/26

```
head(runner)
```

```
#=====
```

```
# 1. Cartesian Plot
```

```
#=====
```

```
p1 <- ggplot(runner,  
             aes(x = Distance,  
                 y = Speed,  
                 color = Speed)) +  
  geom_line(size = 1) +  
  scale_color_gradient(  
    low = "blue",  
    high = "red"  
  ) +  
  ggtitle("Runner Speed vs Distance") +  
  xlab("Distance (km)") +  
  ylab("Speed (km/h)") +  
  theme_minimal()
```

```
print(p1)
```

```
#=====
```

```
# 2. Polar Coordinate Plot
```

```
#=====
```

```
p2 <- ggplot(runner,  
             aes(x = factor(round(Direction/30)*30),  
                 fill = Speed)) +  
  geom_bar() +  
  coord_polar() +
```

R CODE CLASS ACTIVITY -15/7/26

```
ggtitle("Running Direction (Polar Plot)") +  
theme_minimal()
```

```
print(p2)
```

```
#=====
```

```
# 3. Heart Rate Plot
```

```
#=====
```

```
p3 <- ggplot(runner,  
             aes(x = Distance,  
                 y = HeartRate,  
                 color = HeartRate)) +  
geom_line(size = 1) +  
scale_color_gradient(  
  low = "green",  
  high = "red"  
) +  
ggtitle("Heart Rate During Marathon") +  
theme_minimal()
```

```
print(p3)
```

```
#=====
```

```
# 4. Elevation Plot
```

```
#=====
```

```
p4 <- ggplot(runner,  
             aes(x = Distance,  
                 y = Elevation)) +  
geom_line(color = "darkgreen") +  
ggtitle("Elevation Profile") +
```

R CODE CLASS ACTIVITY -15/7/26

```
xlab("Distance (km)") +
```

```
ylab("Elevation (m)") +
```

```
theme_minimal()
```

```
print(p4)
```

```
#=====
```

```
# End of Program
```

```
#=====
```

R CODE CLASS ACTIVITY -15/7/26

```
1 #-----
2 # EXPERIMENT 10|
3 #
4 #-----
5
6 # Install packages (Run only once)
7 if (!require(ggplot2)) install.packages("ggplot2")
8 library(ggplot2)
9
10 #-----
11 # Generate Sample Data
12 #-----
13
14 set.seed(123)
15
16 #----- 1000 -----
17 #-----
```

```
+ low = "white",
+ high = "black"
+ ) +
+ ggtitle("Color-Blind Friendly Dashboard") +
+ theme_minimal() +
+ theme(axis.text.x = element_text(angle = 90))
>
> print(p5)
> #=====
> # End of Program
> #=====
>
```

Environment History Connections Tutorial

R - Global Environment

vaccination	360 obs. of 4 variables
vegetation	1000 obs. of 4 variables
wind	1000 obs. of 2 variables

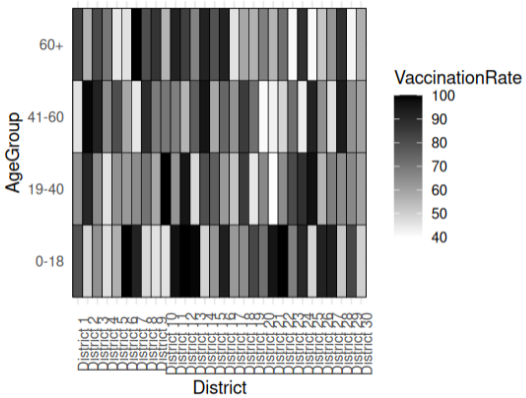
Values

Aftershock	int [1:1000] 17 53 23 8 40 0 56 33 4...
AgeGroup	chr [1:4] "0-18" "19-40" "41-60" "60..."

Files Plots Packages Help Viewer Presentation

Zoom Export

Color-Blind Friendly Dashboard



VaccinationRate

AgeGroup

District

```
1 #-----
2 # EXPERIMENT 9|
3 #
4 #-----
5
6 # Install packages (Run only once)
7 if (!require(ggplot2)) install.packages("ggplot2")
8 library(ggplot2)
9
10 #-----
11 # Generate Sample Data
12 #-----
13
14 set.seed(123)
15
16 #----- 1000 -----
17 #-----
```

```
+ y = Elevation)) +
+ geom_line(color = "darkgreen") +
+ ggtitle("Elevation Profile") +
+ xlab("Distance (km)") +
+ ylab("Elevation (m)") +
+ theme_minimal()
>
> print(p4)
> #=====
> # End of Program
> #=====
>
```

Environment History Connections Tutorial

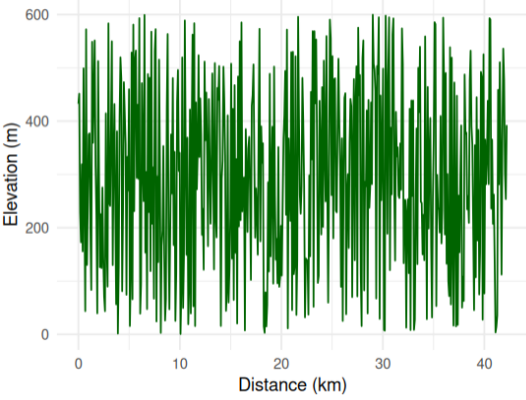
R - Global Environment

runner	500 obs. of 5 variables
sales	1500 obs. of 3 variables
total_sales	50 obs. of 2 variables
vegetation	1000 obs. of 4 variables
wind	1000 obs. of 2 variables

Files Plots Packages Help Viewer Presentation

Zoom Export

Elevation Profile



Elevation (m)

Distance (km)

10)

R CODE CLASS ACTIVITY -15/7/26

```
#=====
```

```
# EXPERIMENT 10
```

```
# COVID-19 Vaccination Campaign Dashboard
```

```
#=====
```

```
# Install packages (Run only once)
```

```
if (!require(ggplot2)) install.packages("ggplot2")
```

```
library(ggplot2)
```

```
#-----
```

```
# Generate Sample Data
```

```
#-----
```

```
set.seed(123)
```

```
District <- paste("District", 1:30)
```

```
AgeGroup <- c("0-18", "19-40", "41-60", "60+")
```

```
Vaccine <- c("Covishield", "Covaxin", "Sputnik")
```

```
vaccination <- expand.grid(
```

```
  District = District,
```

```
  AgeGroup = AgeGroup,
```

```
  Vaccine = Vaccine
```

```
)
```

```
vaccination$VaccinationRate <- sample(40:100,
```

```
  nrow(vaccination),
```

```
  replace = TRUE)
```

R CODE CLASS ACTIVITY -15/7/26

```
head(vaccination)
```

```
#=====
```

```
# 1. Heatmap for Policymakers
```

```
#=====
```

```
p1 <- ggplot(vaccination,  
             aes(x = District,  
                 y = AgeGroup,  
                 fill = VaccinationRate)) +  
  geom_tile() +  
  scale_fill_gradient(  
    low = "yellow",  
    high = "darkgreen"  
  ) +  
  ggtitle("COVID-19 Vaccination Rates (Policy Dashboard)") +  
  xlab("District") +  
  ylab("Age Group") +  
  theme_minimal() +  
  theme(axis.text.x = element_text(angle = 90))  
  
print(p1)
```

```
#=====
```

```
# 2. Public Dashboard - Bar Chart
```

```
#=====
```

```
district_total <- aggregate(  
  VaccinationRate ~ District,
```

R CODE CLASS ACTIVITY -15/7/26

```
vaccination,  
mean  
)
```

```
p2 <- ggplot(district_total,  
             aes(x = reorder(District, VaccinationRate),  
                 y = VaccinationRate,  
                 fill = VaccinationRate)) +  
  geom_col() +  
  coord_flip() +  
  scale_fill_gradient(  
    low = "lightblue",  
    high = "blue"  
  ) +  
  ggtitle("Average Vaccination Rate by District") +  
  xlab("District") +  
  ylab("Average Vaccination Rate (%)") +  
  theme_minimal()
```

```
print(p2)
```

```
#=====
```

```
# 3. Vaccine-wise Box Plot
```

```
#=====
```

```
p3 <- ggplot(vaccination,  
             aes(x = Vaccine,  
                 y = VaccinationRate,  
                 fill = Vaccine)) +
```

R CODE CLASS ACTIVITY -15/7/26

```
geom_boxplot() +  
ggtitle("Vaccination Rate by Vaccine Type") +  
theme_minimal()
```

```
print(p3)
```

```
#=====
```

```
# 4. Age Group Comparison
```

```
#=====
```

```
age_total <- aggregate(  
  VaccinationRate ~ AgeGroup,  
  vaccination,  
  mean  
)
```

```
p4 <- ggplot(age_total,  
  aes(x = AgeGroup,  
    y = VaccinationRate,  
    fill = AgeGroup)) +  
  geom_bar(stat = "identity") +  
  ggtitle("Average Vaccination Rate by Age Group") +  
  xlab("Age Group") +  
  ylab("Average Vaccination Rate (%)") +  
  theme_minimal()
```

```
print(p4)
```

```
#=====
```

R CODE CLASS ACTIVITY -15/7/26

5. Accessibility Version

#=====

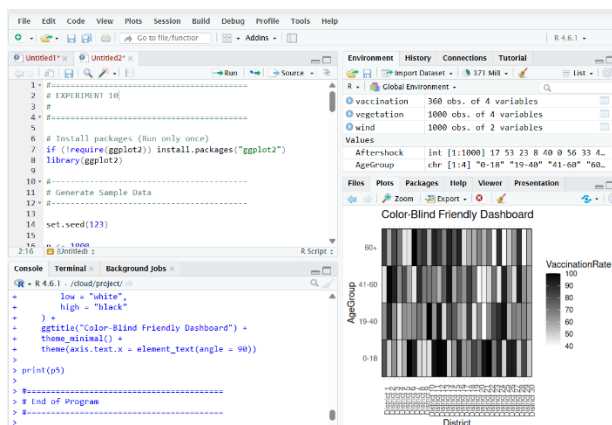
```
p5 <- ggplot(vaccination,
             aes(x = District,
                 y = AgeGroup,
                 fill = VaccinationRate)) +
  geom_tile(color = "black") +
  scale_fill_gradient(
    low = "white",
    high = "black"
  ) +
  ggtitle("Color-Blind Friendly Dashboard") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90))

print(p5)
```

#=====

End of Program

#=====



R CODE CLASS ACTIVITY -15/7/26